

Introduction to Microeconomics
Lecture 2—Comparative Advantage

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Comparative Advantage: The Basis of Exchange

Why do people exchange goods and services in the first place? Why not just produce our own food, cars, clothing, shelter, and the like? The answer is that we can all have more of every good and service if we specialize in the activities at which we are **relatively** most efficient.

Comparative Advantage: The Basis of Exchange

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How much would it cost to produce a chicken sandwich from scratch?

According to this YouTuber^a, it took him 6 months of his life and \$1500 to make a sandwich from scratch! [▶ Link](#)

⚠ Viewer discretion advised! The video shows the murder of two chicken. 😬

^aYou should always be careful with statements said by social media creators. Some of them are knowledgeable and produce good content, but many of them are very ignorant people (that is why they became social media creators in the first place) whose only incentives are to attract the largest number of views by creating click-bait trash content. Use your critical thinking skills before believing what you hear and see in social media!

Ali and Sultan

Ali is a house painter whose room ^f needs fixing, while Sultan is a roofer whose house needs painting. Although Ali is a house painter, he also knows how to repair roofs. Sultan, for his part, knows how to paint houses.

	Painting	Roofing
Ali	300 hours	400 hours
Sultan	200 hours	100 hours

- Note that Sultan has an **absolute advantage** → in each activity, Sultan takes less hours than Ali to perform each task.
- Should Sultan do both the roofing and painting jobs for both houses?

Ali and Sultan

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- Sultan may say that him doing both jobs is unfair. A more telling objection is that it is **inefficient**.

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- The reason is that Ali has a **comparative advantage** over Sultan at painting, which means that he is **relatively** more efficient at painting than Sultan.

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- Key word: **relatively**.

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- It means that Ali has a **lower opportunity cost** at performing the task of painting than Sultan has.

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- Ali: the opportunity cost of painting a house is 0.75 roofing jobs.

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- It means that Ali has a **lower opportunity cost** at performing the task of painting than Sultan has.
- Ali: the opportunity cost of painting a house is 0.75 roofing jobs.
- Sultan: the opportunity cost of painting a house is 2 roofing jobs.

$$\text{Ali : } 1 \text{ ~~house~~ }^{\text{paint}} \left(\frac{300 \text{ ~~hours~~}}{\text{~~house~~}} \right) \left(\frac{\text{roof}}{400 \text{ ~~hours~~}} \right) = \frac{300}{400} = \frac{3}{4} \text{ roofs} = 0.75 \text{ roofs}$$

$$\text{Sultan : } 1 \text{ ~~house~~ }^{\text{paint}} \left(\frac{200 \text{ ~~hours~~}}{\text{~~house~~ ~~point~~}} \right) \left(\frac{\text{roof}}{100 \text{ ~~hours~~}} \right) = 2 \text{ roofs}$$

$$\text{Ali : C.O. (roofing)} = \frac{4}{3} \text{ paint jobs}$$

$$\text{Sultan : C.O. (roofing)} = \frac{1}{2} \text{ paint jobs}$$

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- Key word: **relatively**.
- It means that Ali has a **lower opportunity cost** at performing the task of painting than Sultan has.
- Ali: the opportunity cost of painting a house is 0.75 roofing jobs.
- Sultan: the opportunity cost of painting a house is 2 roofing jobs.
- Ali has a comparative advantage at painting, and it makes sense for him to do both painting jobs, leaving the two roofing jobs to Sultan.

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- It means that Ali has a **lower opportunity cost** at performing the task of painting than Sultan has.
- Ali: the opportunity cost of painting a house is 0.75 roofing jobs.
- Sultan: the opportunity cost of painting a house is 2 roofing jobs.
- Ali has a comparative advantage at painting, and it makes sense for him to do both painting jobs, leaving the two roofing jobs to Sultan.
- If each performed the painting and roofing task, the time spent would be 700 for Ali and 300 for Sultan. By **specializing**, the total work time falls to 600 for Ali and 200 for Sultan, saving 100 hours each.

The principle of Comparative Advantage

Everyone does best when each person (or country) concentrates on the activities in which he or she is relatively most efficient.

Caution!

Pay close attention to the form in which the productivity information is provided. Your goal in each case is to find each person's opportunity cost of producing the good in question.

Aiym and Balzhan

Aiym can milk a goat in 10 minutes and shear a sheep in 4 minutes. Balzhan can milk a goat in 6 minutes and shear a sheep in 3 minutes. Which statement below is true?

- ① Aiym has no comparative advantage.
- ② Balzhan should do both tasks because he has an absolute advantage in both.
- ③ Aiym has a comparative advantage in shearing sheep and Balzhan has a comparative advantage in milking goats.
- ④ Aiym has a comparative advantage in milking goats and Balzhan has a comparative advantage in shearing sheep.
- ⑤ None of the above statements is true.

Aiyem and Balzhan

Aiyem can milk a goat in 10 minutes and shear a sheep in 4 minutes. Balzhan can milk a goat in 6 minutes and shear a sheep in 3 minutes. Which statement below is true?

C.O. (milking goats)

Aiyem $1 \cancel{\text{goat}} \left(10 \frac{\text{min}}{\cancel{\text{goat}}} \right) \left(1 \frac{\cancel{\text{sheep}}}{\text{min}} \right) = \frac{5}{2} \text{ sheep} = 2.5 \text{ sheep} \rightarrow \text{C.O.}^A(\text{sheep}) = \frac{2}{5} \text{ goats}$

Balzhan $1 \cancel{\text{goat}} \left(6 \frac{\text{min}}{\cancel{\text{goat}}} \right) \left(3 \frac{\cancel{\text{sheep}}}{\text{min}} \right) = 2 \text{ sheep} \rightarrow \text{C.O.}^B(\text{sheep}) = \frac{1}{2} \text{ goats}$

Aiym and Balzhan

Aiym can milk 10 goats per hour or shear 4 sheep per hour. Balzhan can milk 6 goats per hour or shear 3 sheep per hour. Which statement below is true?

- ① Aiym has no comparative advantage.
- ② Balzhan should do both tasks because he has an absolute advantage in both.
- ③ Aiym has a comparative advantage in shearing sheep and Balzhan has a comparative advantage in milking goats.
- ④ Aiym has a comparative advantage in milking goats and Balzhan has a comparative advantage in shearing sheep.
- ⑤ None of the above statements is true.

Aiyun can milk 10 goats per hour or shear 4 sheep per hour. Balzhan can milk 6 goats per hour or shear 3 sheep per hour. Which statement below is true?

C.O. (of goat milking)

Aiyun

$$1 \cancel{\text{goat}} \left(10 \frac{\text{hour}}{\cancel{\text{goats}}} \right) \left(4 \frac{\text{sheep}}{\text{hour}} \right) = \frac{2}{5} \text{ sheep}$$

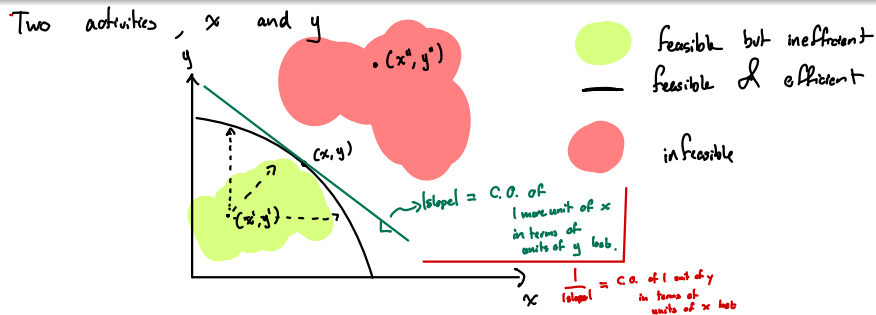
Balzhan

$$1 \cancel{\text{goat}} \left(6 \frac{\text{hour}}{\cancel{\text{goats}}} \right) \left(3 \frac{\text{sheep}}{\text{hour}} \right) = \frac{1}{2} \text{ sheep}$$

The production possibilities frontier (PPF)

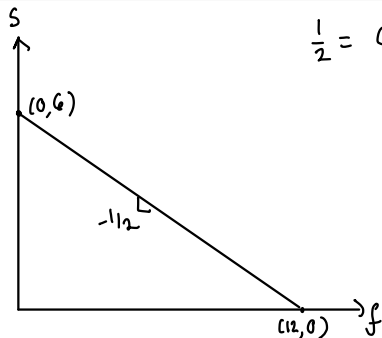
Production Possibilities Frontier

It is the boundary between those combinations of goods and services that can be produced, given current resources, and those that cannot. Any point inside the PPF is **feasible** but **inefficient**. Any point outside the PPF is **infeasible**. Points on the frontier are **efficient**. The slope of the PPF determines the **opportunity cost** of one activity/good vs the other one.



Daniyar and the island

Daniyar has been cast away to a remote island, where he is the only inhabitant. Daniyar can produce 6 m^2 /week of shelter or 12 kg/week of food. Describe Daniyar's PPF.



$\frac{1}{2} = C.O.(\text{food}) :=$ cost of opp. of 1 unit of food
in terms of units of shelter

"for every extra kg of food, Daniyar gives up
0.5 m^2 of shelter"

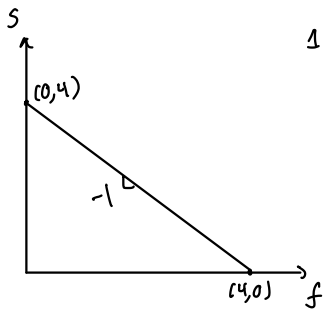
(Using (0,6) with the slope-point formula)

$$S - 6 = -\frac{1}{2}(f - 0) \Rightarrow S = 6 - \frac{1}{2}f$$

$$\hookrightarrow f = 12 - 2S$$

Abuzer

Abuzer can produce 4 m^2 /week of shelter or 4 kg/week of food. Describe Abuzer's PPF.



1 = c.o. of f in terms of S

$$S = 4 - f$$

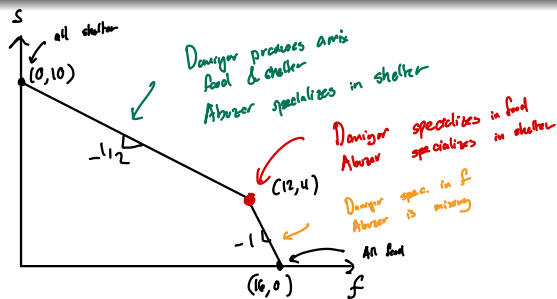
$$f = 4 - S$$

PPF in a two-person Economy

In the examples above, Daniyar has a comparative advantage in producing food, while Abuzer has a comparative advantage producing shelter.

Daniyar and Abuzer

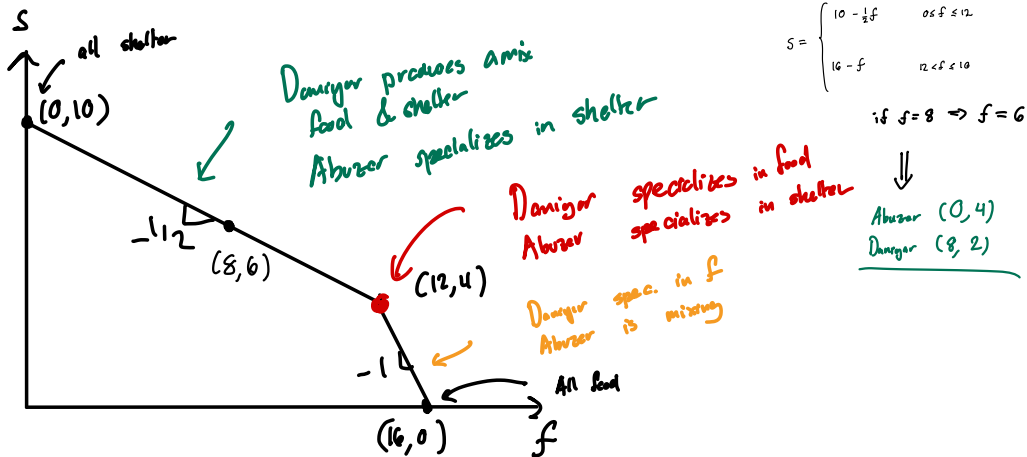
Daniyar can produce 6 m^2 /week of shelter and 12kg/week of food. Abuzer can produce 4 m^2 /week of shelter and 4 kg/week of food. Describe the economy's PPF.



$$S = \begin{cases} 10 - \frac{1}{2}f & 0 \leq f \leq 12 \\ 16 - f & 12 < f \leq 16 \end{cases}$$

Daniyar and Abuzer

Suppose Daniyar and Abuzer decide to join forces and have decided to consume, jointly, 6 m^2 /week of shelter and 8 kg/week of food. How should they divide the task of producing these quantities?

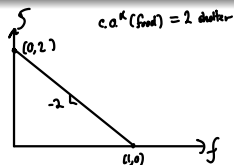


The Principle of Increasing Opportunity Cost

In expanding the production of any good, first employ those resources with the lowest opportunity cost, and only afterward turn to resources with higher opportunity costs.

Kamilla joins the group of castaways

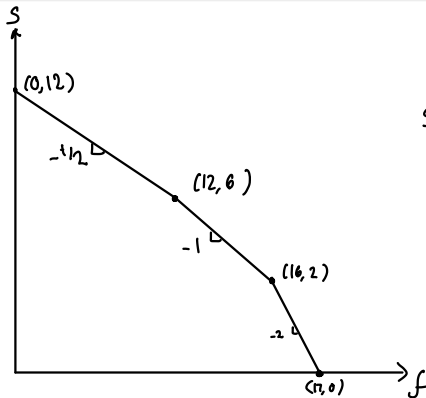
Suppose that Kamilla also becomes stranded in the island. Kamilla can produce $2 m^2$ /week of shelter or 1 kg/week of food. Start by describing her individual PPF. Then describe the PPF of the three-person economy. If they jointly decide to produce 14 kg/week of food and $4 m^2$ /week of shelter, how should each person's work time be allocated?



$c.o.^K(food) = 2 \text{ shelter}$

C.O. (food)

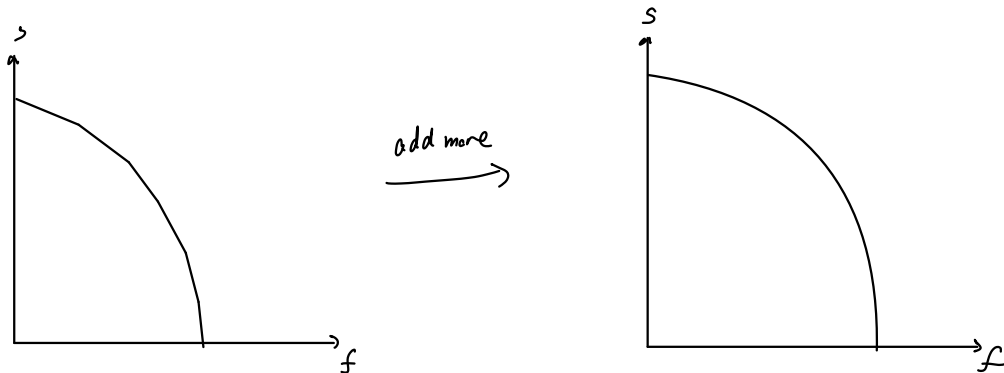
Danigar	1/2
Abuser	1
Kamilla	2
D	12 f
A	4 f
K	1 f



$$S = \begin{cases} 12 - \frac{1}{2}f & 0 \leq f \leq 12 \\ 16 - f & 12 < f \leq 16 \\ 34 - 2f & 16 < f \leq 17 \end{cases}$$

The production possibilities frontier with many workers

Intuition behind its shape: produce the initial units of clothing using the resources that are relatively most efficient at clothing production, and only then turn to those that are relatively less efficient at clothing production.



Another example of a bowed out (strictly concave) PPF

Suppose that there are two goods, x and y , which need labor as an input of production. Let l_x and l_y be the amounts of labor used by the production of x and y respectively. Suppose that there are 100 hours of labor available. Assume also that the production process of x and y can be described by:

$$x = \sqrt{l_x}, \quad y = \sqrt{l_y}.$$

Describe, mathematically and graphically the PPF.